

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Date:

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

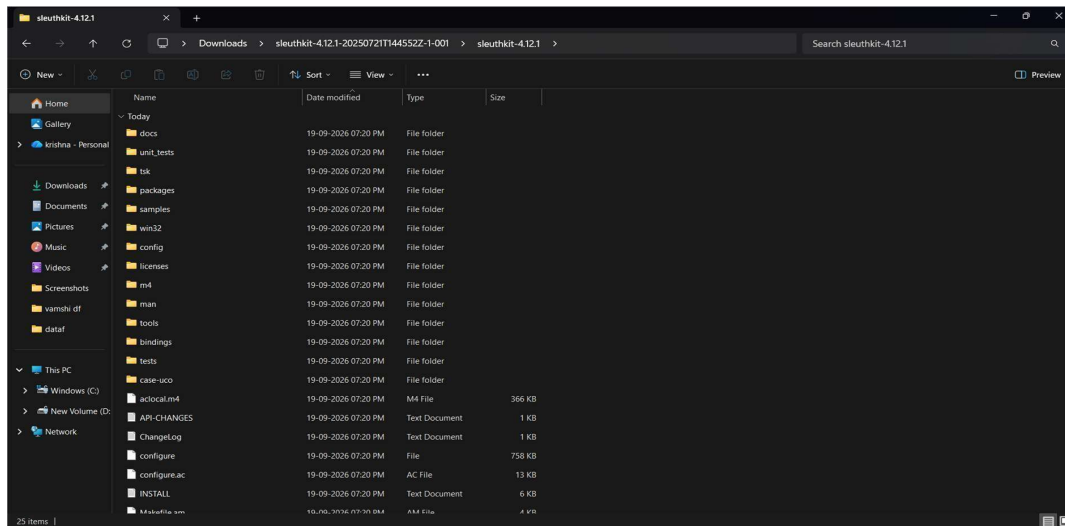
Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.



Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

 4Dell Latitude CPi.E01	19-09-2026 07:23 PM	E01 File	6,55,366 KB
 4Dell Latitude CPi.E02	19-09-2026 07:23 PM	E02 File	4,09,556 KB

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

fsstat [image file] > filesystem_info.txt

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

```
C:\Users\krish\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -i ewf -o 63 "C:\Users\krish\Downloads\4Dell Latitude CPi.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2897152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
```

3. List Partitions with mmls:

- Run the command:

arduino

mmls [image file] > partitions.txt

- This command lists the partitions within the image file.

```
C:\Users\krish\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls -i ewf "C:\Users\krish\Downloads\del\4Dell Latitude CPl.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000000062	0000000063	Unallocated
002:	000:000	0000000063	0009510479	0009510417	NTFS / exFAT (0x07)
003:	-----	0009510480	0009514259	0000003780	Unallocated

4. Analyze File System with fls:

- Run the command:

arduino

fls -r [image file] > file_list.txt

- This command recursively lists files and directories in the file system, showing their metadata.

```
C:\Users\krish\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 -r "C:\Users\krish\Downloads\del\4Dell Latitude CPl.E01"
d/d 3671-144-7: Documents and Settings
+ d/d 3673-144-7: All Users
++ d/d 3707-144-7: Application Data
+++ r/r 3800-128-1: desktop.ini
+++ d/d 3708-144-7: Microsoft
++++ d/d 3709-144-1: Crypto
+++++ d/d 3712-144-1: DSS
+++++ d/d 3713-144-1: MachineKeys
+++++ d/d 3710-144-1: RSA
+++++ d/d 3711-144-1: MachineKeys
+++++ d/d 10977-144-1: Dr Watson
+++++ r/r 10979-128-3: drwtsn32.log
+++++ r/r 10980-128-3: user.dmp
++++ d/d 7325-144-1: HTML Help
++++ d/d 7287-144-7: Media Index
+++++ r/r 6074-128-5: wmpLibrary_v_0_12.db
+++++ d/d 4759-144-1: Network
```

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

icat [image file] [inode number] > [output file]

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

istat [image file] [inode number] > metadata_info.txt

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

fls -m / -r [image file] > body.txt

- Then create the timeline:

CSS

mactime -b body.txt > timeline.txt

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

```
C:\Users\krish\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -i ewf -o 63 "C:\Users\krish\Downloads\delldell\Latitude CPi.E01" 10979
MFT Entry Header Values:
Entry: 10979          Sequence: 2
LogFile Sequence Number: 36691948
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 330 (S-1-5-21-2000478354-688789044-1708537768-1003)
Created: 2004-08-20 20:55:45.274075200 (India Standard Time)
File Modified: 2004-08-20 20:55:48.278395200 (India Standard Time)
MFT Modified: 2004-08-20 20:55:48.278395200 (India Standard Time)
Accessed: 2004-08-20 20:55:48.278395200 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: drwtsn32.log
Parent MFT Entry: 10977          Sequence: 2
Allocated Size: 0          Actual Size: 0
Created: 2004-08-20 20:55:45.274075200 (India Standard Time)
File Modified: 2004-08-20 20:55:45.274075200 (India Standard Time)
MFT Modified: 2004-08-20 20:55:45.274075200 (India Standard Time)
Accessed: 2004-08-20 20:55:45.274075200 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 90
Type: $DATA (128-3) Name: N/A Non-Resident size: 37044 init_size: 37044
5160328 5160329 5160330 5160331 5160332 5160333 5160334 5160335
5160336 5160337 5160338 5160339 5160340 5160341 5160342 5160343
5160344 5160345 5160346 5160347 5160348 5160349 5160350 5160351
5160352 5160353 5160354 5160355 5160356 5160357 5160358 5160359
5160360 5160361 5160362 5160363 5160364 5160365 5160366 5160367
5160368 5160369 5160370 5160371 5160372 5160373 5160374 5160375
5160376 5160377 5160378 5160379 5160380 5160381 5160382 5160383
5160384 5160385 5160386 5160387 5160388 5160389 5160390 5160391
5160392 5160393 5160394 5160395 5160396 5160397 5160398 5160399
5160400
```

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

 timeline	19-09-2026 09:53 PM	Text Document	0 KB
 body	19-09-2026 09:52 PM	Text Document	3,174 KB
 recovered_drwtstn32	19-09-2026 09:51 PM	Text Document	37 KB
 filesystem_info	19-09-2026 09:13 PM	Text Document	0 KB
 mmls	19-09-2026 08:34 PM	Application	74 KB
 mmstat	19-09-2026 08:34 PM	Application	72 KB
 tsk_comparedir	19-09-2026 08:34 PM	Application	774 KB
 tsk_gettimes	19-09-2026 08:34 PM	Application	887 KB
 tsk_imageinfo	19-09-2026 08:34 PM	Application	889 KB
 tsk_loaddb	19-09-2026 08:34 PM	Application	1,478 KB

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The forensic disk image was successfully analyzed using Sleuth Kit. File-system information and partition details were identified, files and directories were listed, file metadata was examined, selected files could be recovered using their inode numbers, and a timeline of file activity was generated.